

Database Triggers (Self-Learning)

Q. What is a Trigger in SQL? Discuss the different types of triggers (BEFORE and AFTER) and explain their primary uses.

****Definition to Remember****

A ****Trig**
(**INSE**
RDBMS

1. Types of Triggers

Triggers are classified by when they execute relative to the triggering DML event:

A. BEFORE Trigger

* Exec

Example: Before inserting into the `Employee` table, a BEFORE trigger checks if the incoming salary is below minimum wage. If so, it raises an error and aborts the insert.

B. AFTER Trigger

* Exec

Example: After an update to `Employee`, an AFTER trigger inserts a record into `Employee\_Audit` logging the user, timestamp, and what changed.

(Some systems also support INSTEAD OF Triggers - primarily used on Views to enable update operations on non-updatable views.)

2. Trigger Event Classification

| BEFORE | AFTER |

| :--- | :-

3. Uses and Advantages

| Use Case | Explanation |

| :--- | :-

****Must-Write Points (for 10 marks)****

1. A Trig

Quick Recall

`Trigger
Rules, A